

Musical Genre Classifier

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Introduction

- The music industry is a **multi-billion-dollar market** experiencing rapid growth
- New technologies enable anyone to create and distribute music globally
- **Streaming platforms** like Spotify and Amazon Music must categorize tracks to enhance user experience, e.g., by creating personalized, context-aware playlists
- **Automated genre classification** using modern machine learning techniques can significantly speed up this process
- Classifiers such as **Random Forest** and **SVM** often show strong performance on numerical feature datasets, and can be used on musical-specific features to classify each track by its genre

Related Works

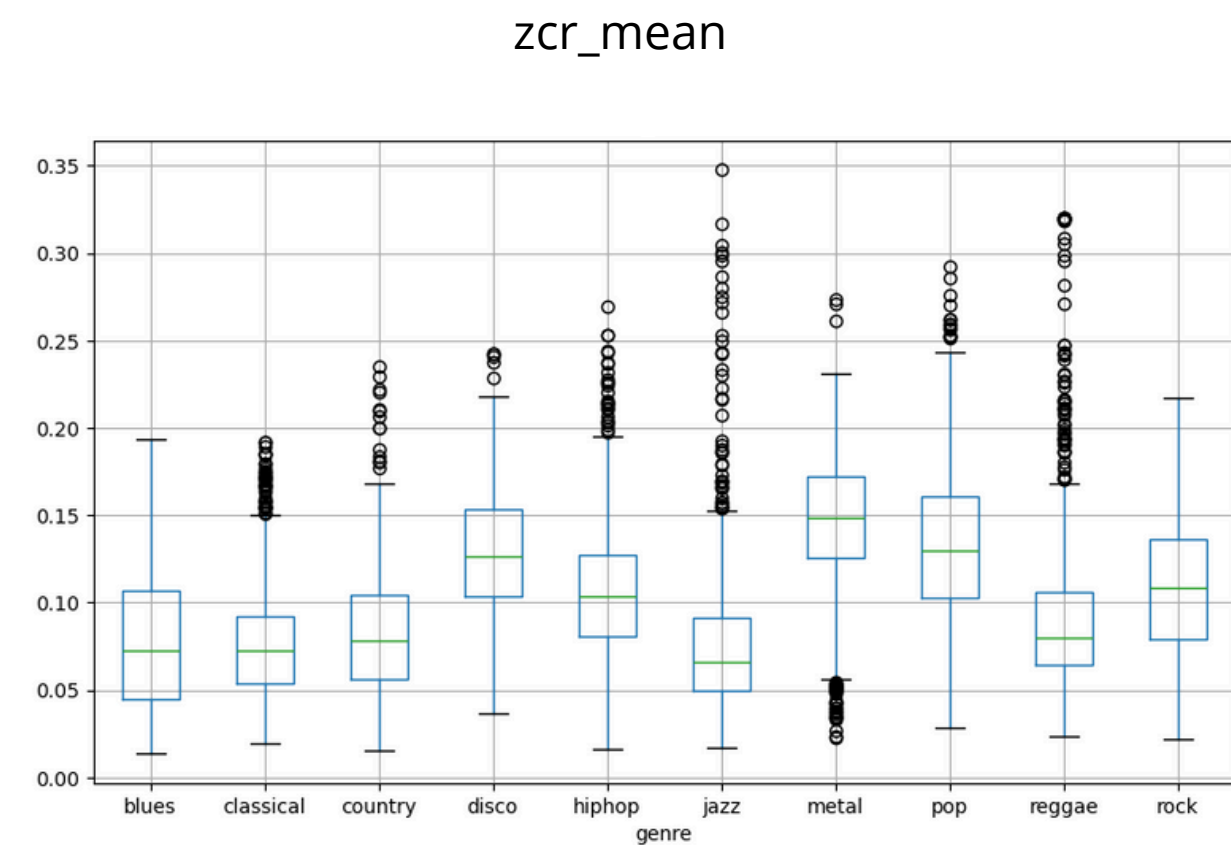
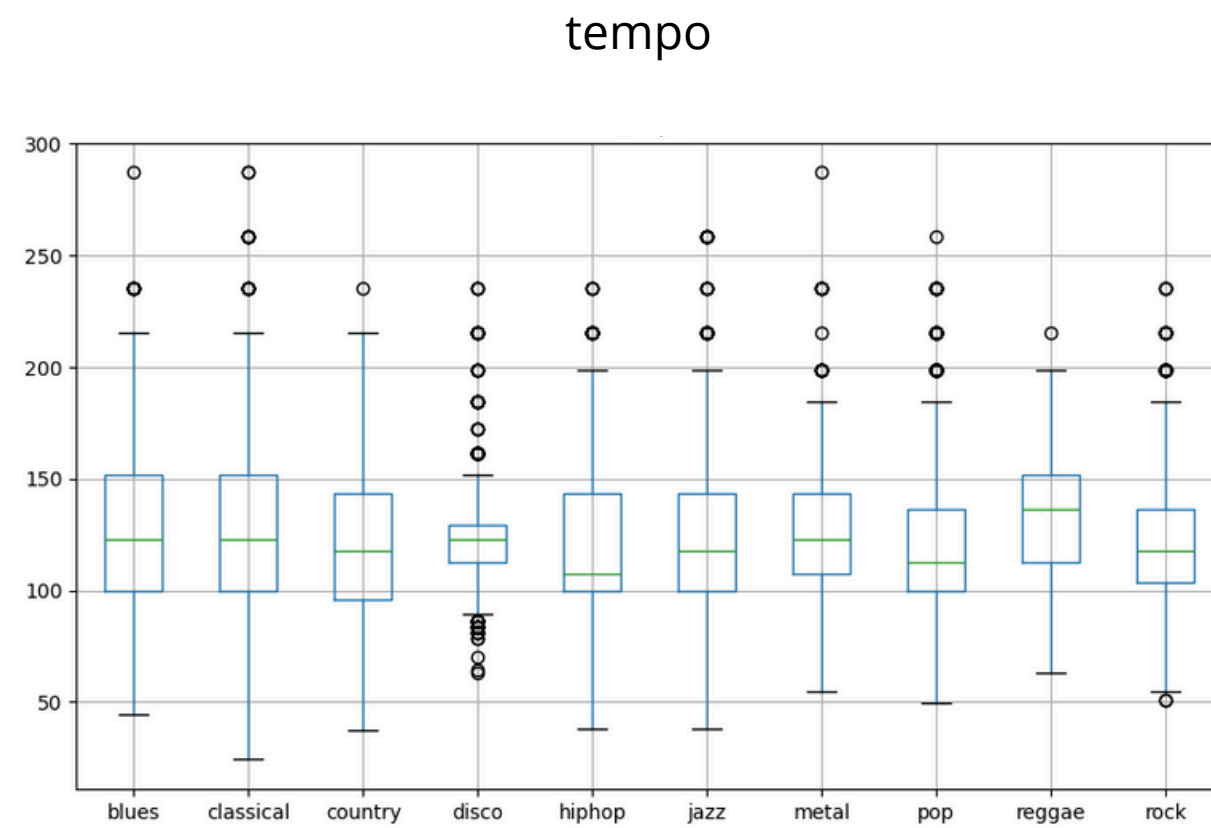
- [1] ***Musical Genre Classification of Audio Signals***, George Tzanetakis, Perry Cook (2002)
 - Authors use Gaussian (GS) classifier, Gaussian mixture model (GMM) classifier and K-nearest neighbor (K-NN) classifier on a smaller dataset of audio files, with about 61% accuracy
- [2] ***A Comparative Study on Content-Based Music Genre***, Tao Li, Mitsunori Ogihara, Qi Li (2003)
 - Authors propose a new feature extraction method based on wavelet (DWCH), increasing the accuracy of [1] to 80%, and use SVM, KNN, GMM and LDA for further analysis
- Both studies involved 30 seconds audio tracks, while I used **3 seconds tracks**
- I used a **RandomForest classifier** (and other classification models), which is not involved in these studies

Dataset

- **GTZAN Dataset - Music Genre Classification**
- 1000 .wav files, each of 30 seconds and classified by their genre (10 genres, 100 files each)
- These files are split into **3s tracks** and analyzed with the Librosa library to extract spectral and musical features
- **55 extracted features:** MFCCs, Chroma, Contrast, spectral centroid, rolloff and bandwidth, ZCR, RMS and tempo
- Reduced input size: **1.3GB → 6MB**

Data Exploration

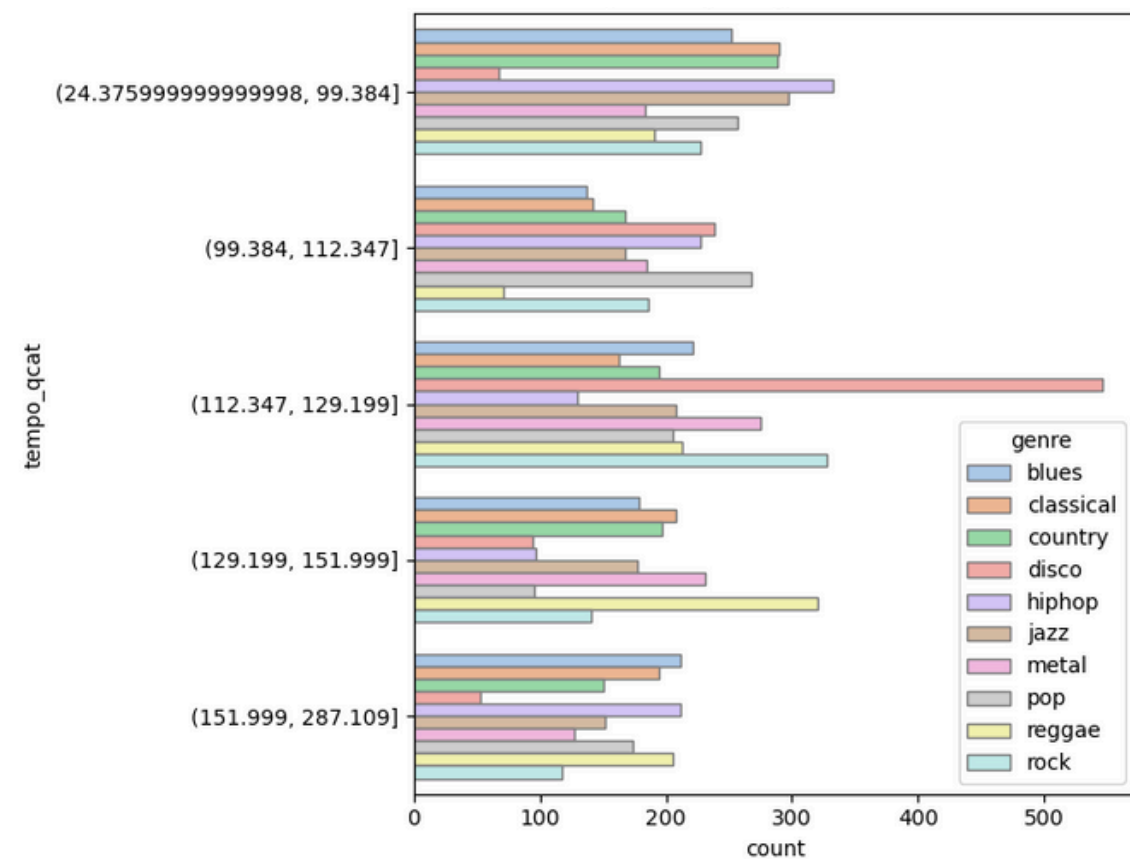
- The dataset is **well-balanced**
- **Boxplots** show the distribution of values in the “tempo” and “zcr_mean” features



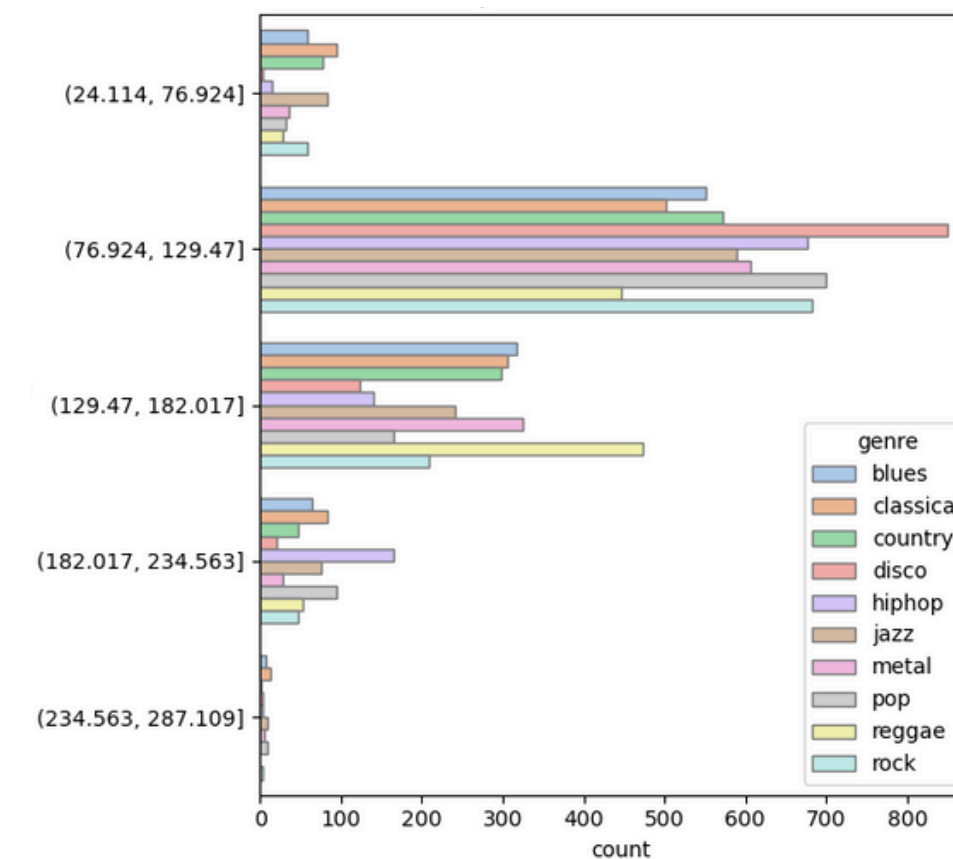
Data Exploration

- **Discretization** can show whether it is possible to classify a musical genre simply by observing the interval in which the feature falls.

Equal-frequency Discretization



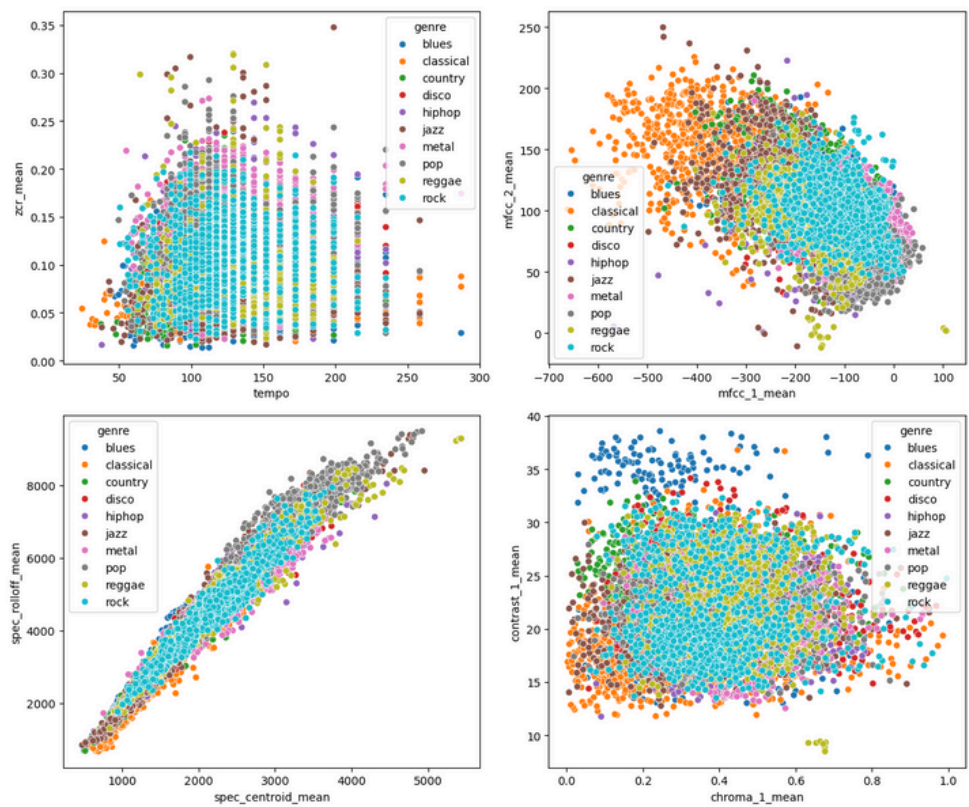
Equal-width Discretization



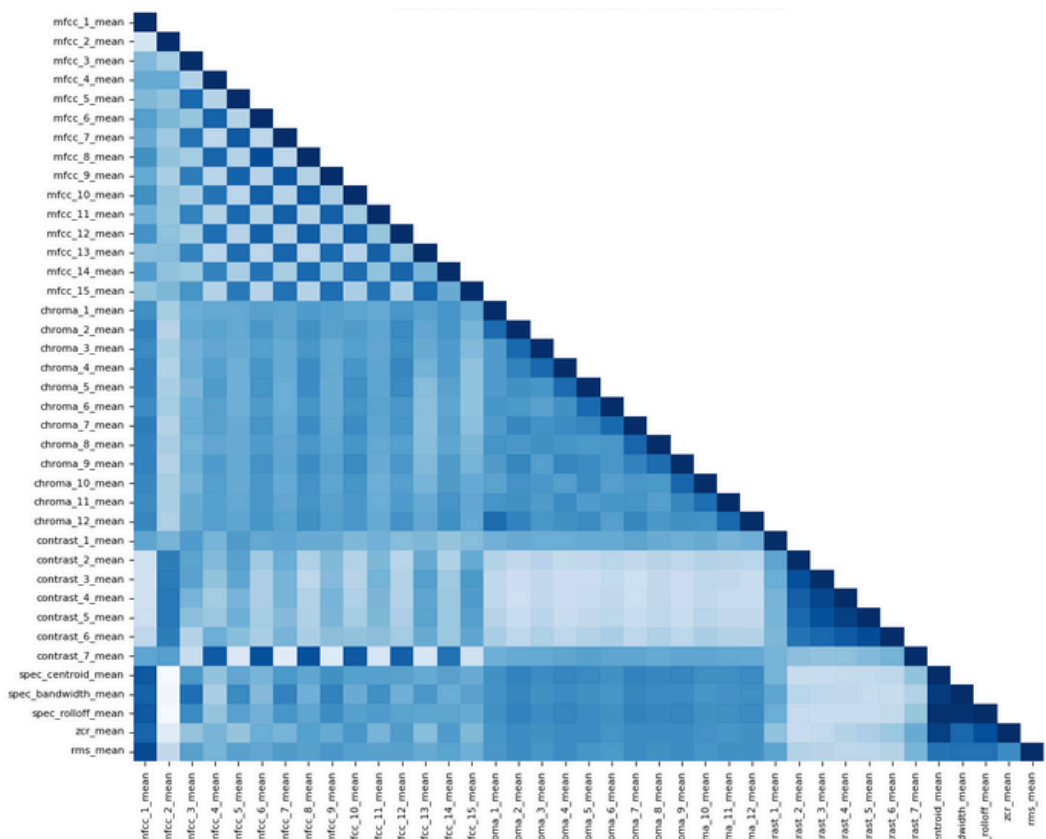
Data Exploration

- **Correlation** between features can help identify the most redundant ones

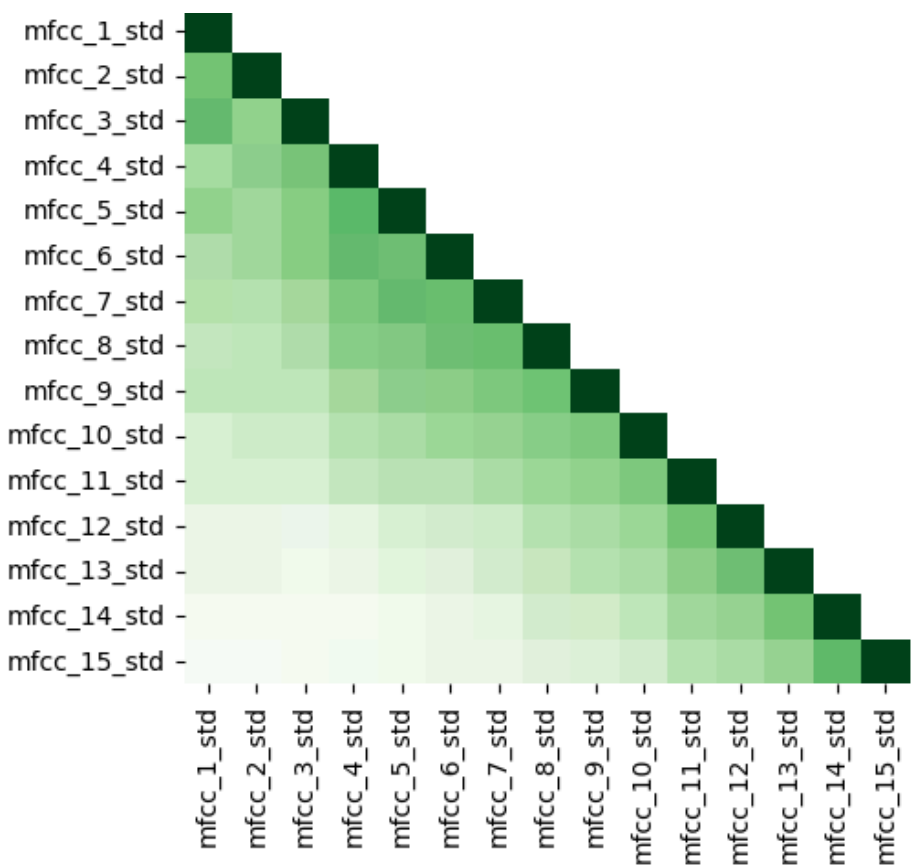
Feature pairs scatterplots



Correlation matrix on “mean” features

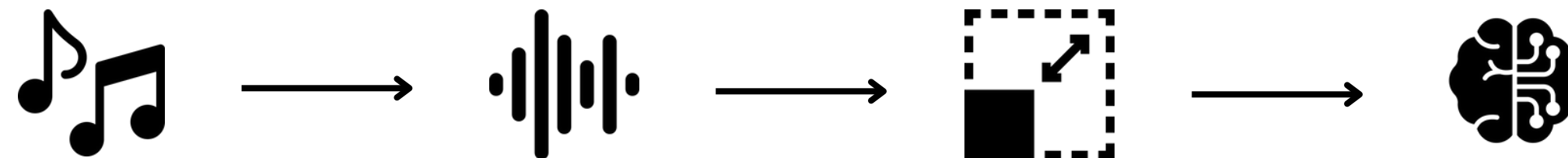


Correlation matrix on “std” features



Data Preprocessing

- Dataset is **fully numerical** except for the “genre” attribute
- No missing values, balanced classes
- Features are processed with **StandardScaler**
 - fit after train/test split to avoid data leakage
- Label encoding is handled internally by Scikit-learn models with the **LabelEncoder**



Data Reduction

- The final goal is to produce **high quality classification** while **reducing the input features** to handle big volumes of data
- Models are trained and tested on 3 different feature sets:
 - **All**: 55 features
 - **Principal Component Analysis** (PCA) with 0.95 cumulative variance: 37 features
 - **Recursive Feature Elimination** (RFE): # of features depends on the model [21-55]

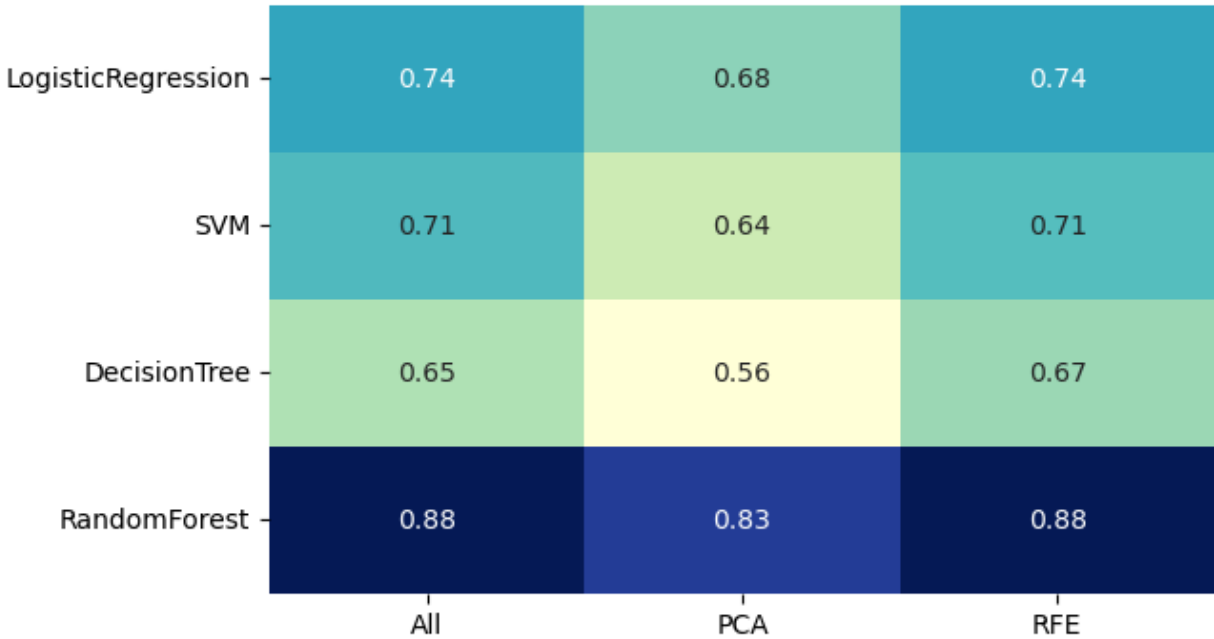
Model Selection

- Models are tested with **GridSearch with cross-validation**: Logistic Regression, LinearSVC, Decision Tree, Random Forest

Parameters grid

Model	Parameter	Values
Logistic Regression	C	{0.1, 1, 10}
	penalty	{l2}
	solver	{lbfgs}
SVM (LinearSVC)	C	{0.1, 1, 10}
Decision Tree	max_depth	{None, 5, 10, 20}
	min_samples_split	{2, 5, 10}
	min_samples_leaf	{1, 2, 4}
Random Forest	n_estimators	{100, 500}
	max_depth	{None, 10, 20}
	min_samples_split	{2, 5}

Accuracy heatmap



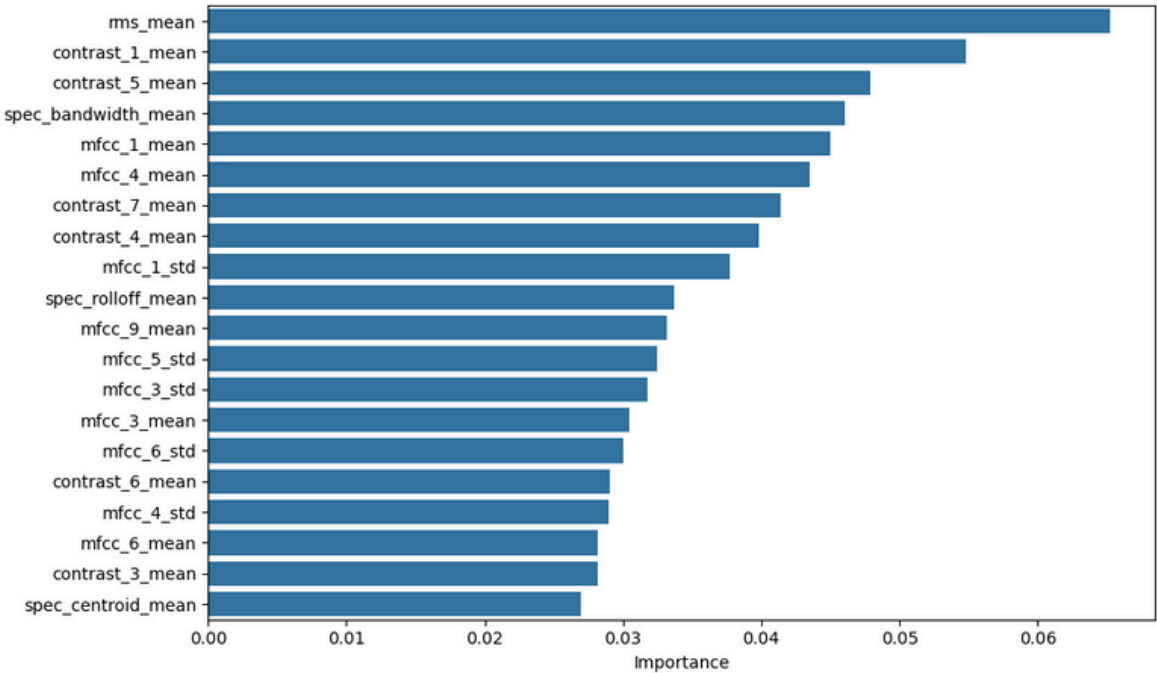
RandomForest Optimization

- The RandomForest model was tested on a bigger grid of parameters to achieve the best possible **classification metrics**

Parameters grid

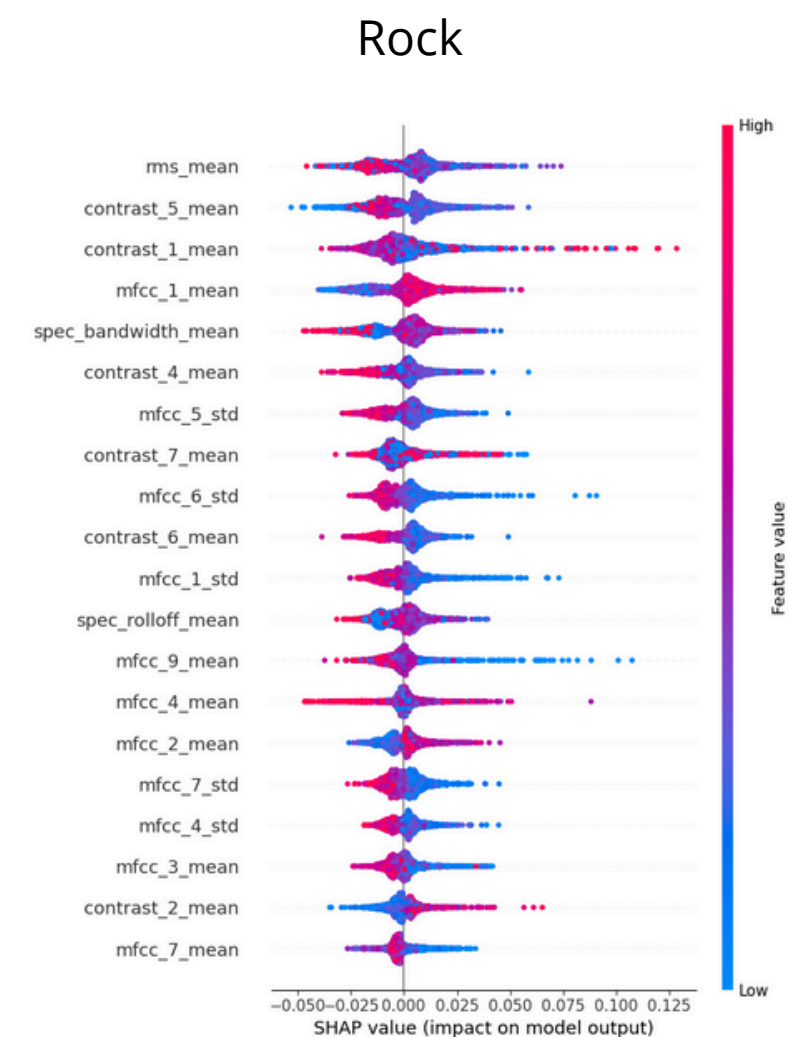
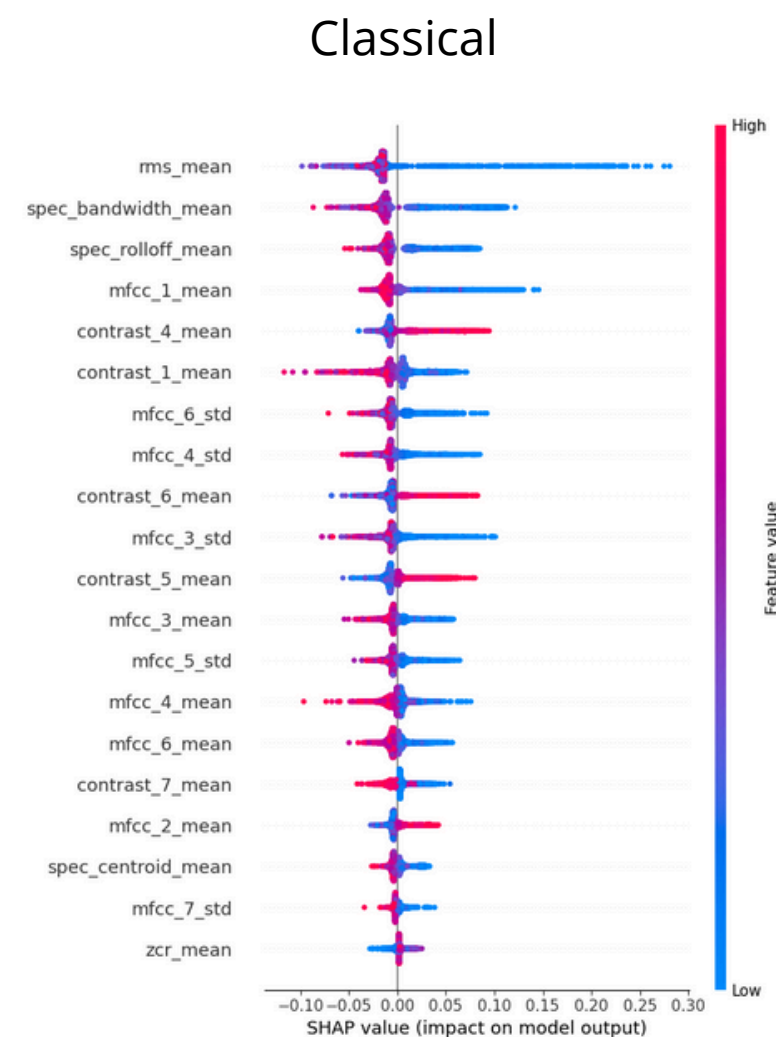
Parameter	Values	Best Model
n_estimators	{100, 500, 1000}	1000
max_depth	{None, 20, 50}	20
min_samples_split	{2, 10, 20}	2
min_samples_leaf	{1, 5, 10}	1
max_features	{sqrt, log2, None}	sqrt
bootstrap	{True, False}	False

Feature Importance



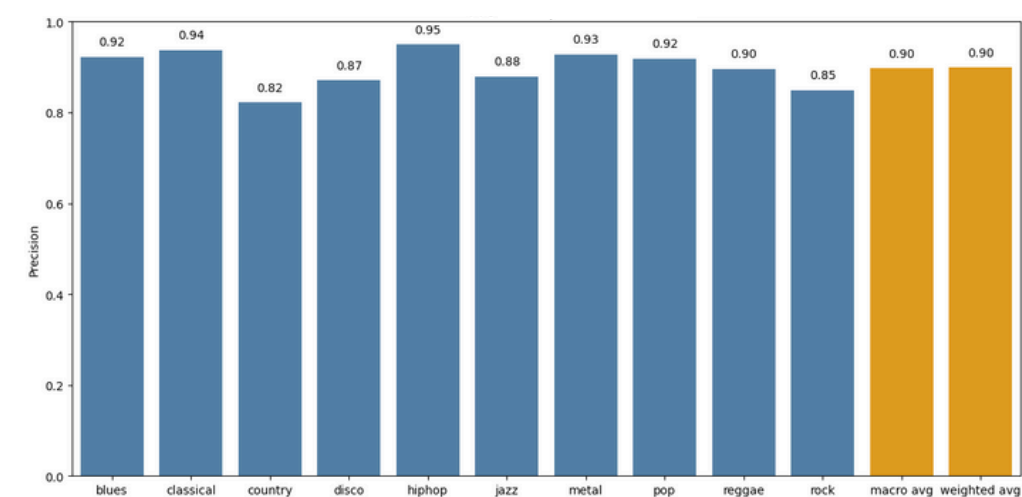
Results

- **SHAP values** help analyze which ranges of feature values is more important in the classification of each class, such as “classical” and “rock”

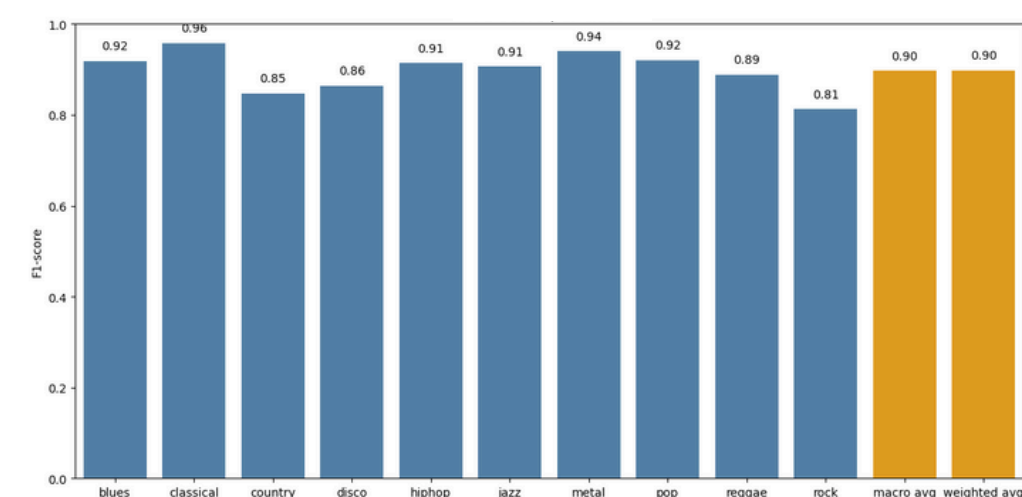


Results

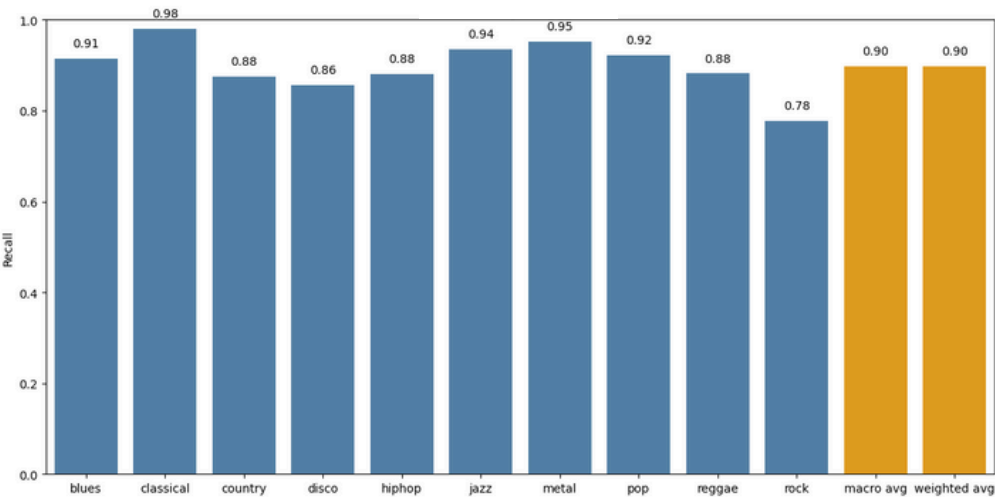
Precision



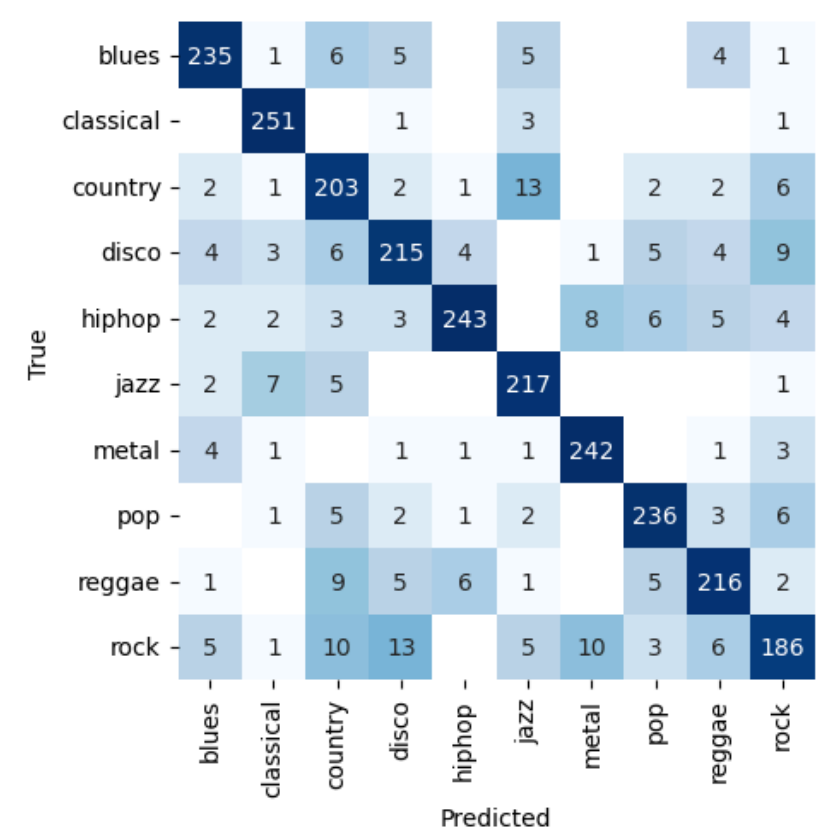
Recall



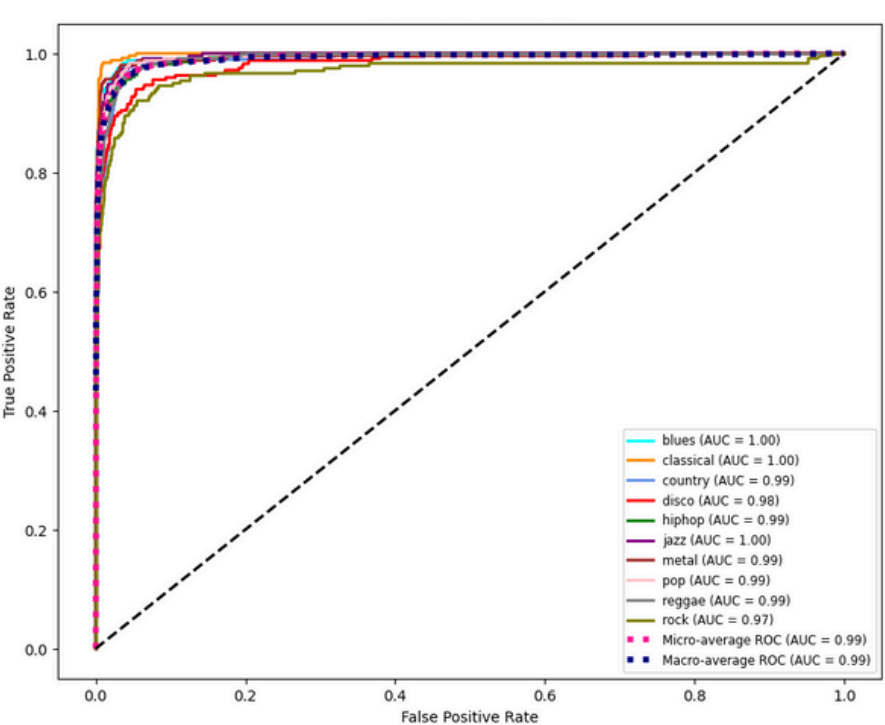
F1-score



Confusion Matrix



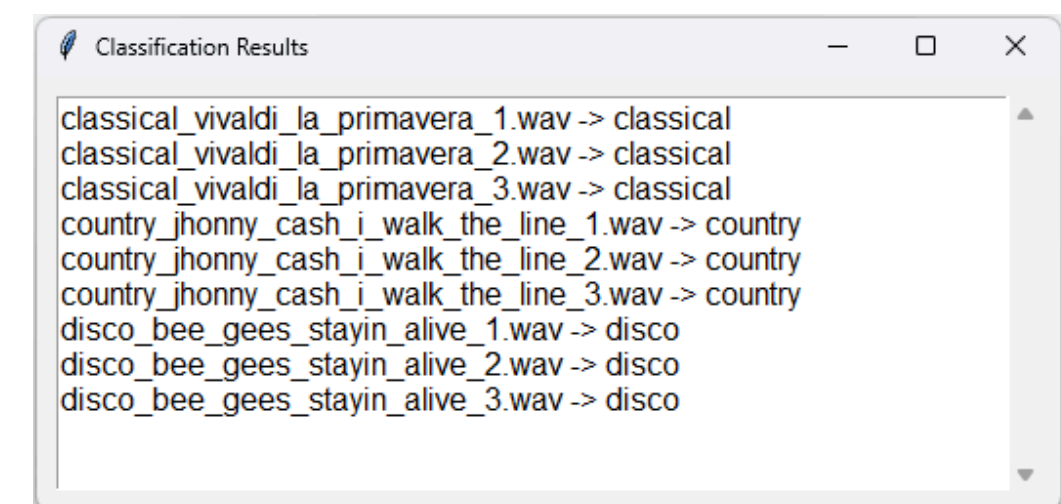
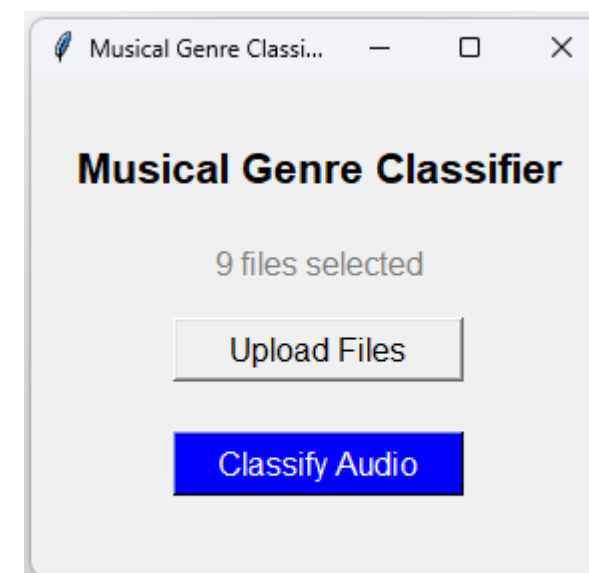
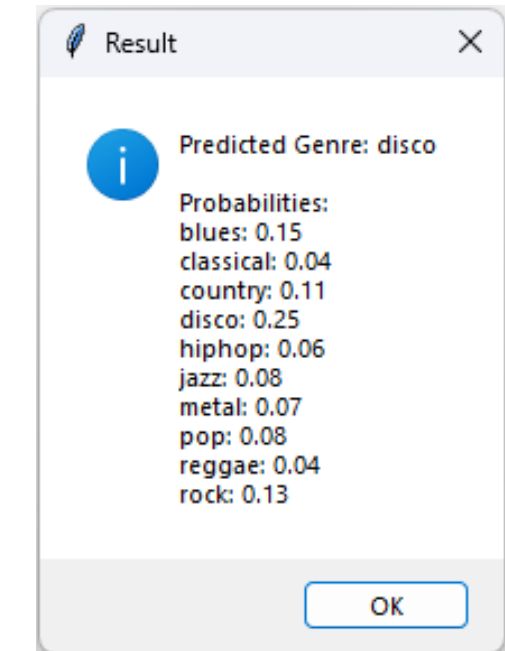
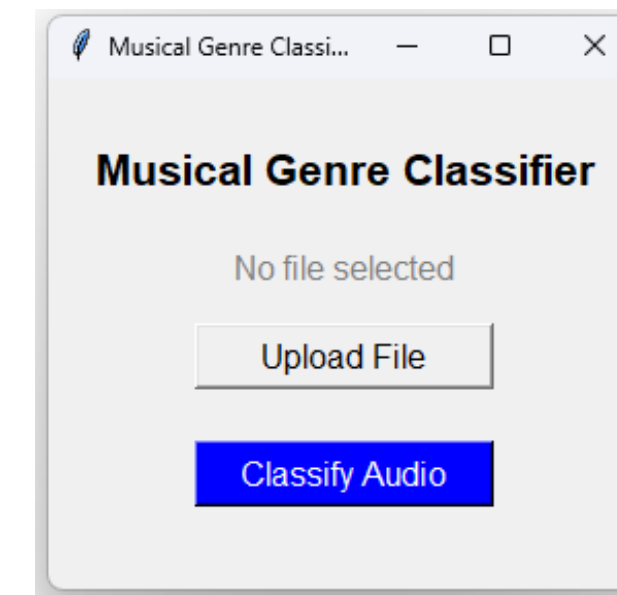
ROC Curve



- **Accuracy:** 90%
- **Micro average AUC:** 0.99
- **Macro average AUC:** 0.99
- **Best:** classical, metal
- **Worst:** rock, hiphop

GUIs

- Two GUIs with slightly different purposes:
 - **Probabilities GUI:** single input file, outputs prediction probabilities
 - **Multifile GUI:** multiple input files, outputs only the classification



Conclusions

- Librosa + Random Forest + RFE → **efficient** and **interpretable** classifier
- **90% accuracy** across 10 genres
- Useful for streaming platforms managing large scale music databases
- **Future work:**
 - Expand the dataset with more modern audio tracks differing from those already present within the same genre, as many tracks are too similar
 - Explore more hyperparameter ranges in the RandomForest optimization

Thank you for your attention